Step2: Add the binary digit generated to the adjacent bit of the Gray code.

1 1 1 0 Gray

1 0 Binary

Step 3: Add the binary digit generated to the adjacent bit of Gray codes.

1 1 1 0 Gray

1 0 1 Binary

Step 4: Add the binary digit generated to the adjacent bit of gray code.

1 1 1 0

1 0 1 1

Hence, $(1110)_G = (1011)_2$

8.7. LOGIC GATES

Logic gates are digital circuits. Digital circuits operate in binary modes, each input and output signal is either '1' or '0'. The '1' and '0' designation represents predefined voltage ranges. These electronic switching circuits are called as logic gates. Each logic gate can have one or more inputs and only one output.

All logic gates can be analysed by constructing a truth table. A truth table represents all possible input and the corresponding output combinations.

The term "logic" is usually used to refer to a decision making process. A logic gate makes logical decisions regarding the existence of output depending upon the nature of the input. Hence, such circuits are called logic circuits.

8.7.1 Basic Logic Gates

The three basic logic gates that makeup all digital circuits are

i) OR gate
ii) AND gate
iii) NOT gate.

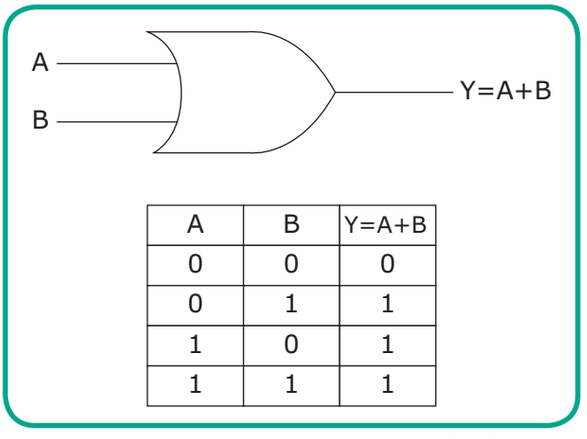
The following points may be noted about logic gates.

1. A binary '0' represents 0V and binary '1' represents +5V. It is common to refer to binary '0' as LOW input or output and binary '1' as HIGH input or output.
2. A logic gate has only one output and the output will depend upon the input signals and the type of gates.
3. The operation of a logic gate may be described either by truth table or Boolean algebra.

8.7.2 OR Gate

An OR gate has two or more input signals and only one output signal. An OR gate performs logical addition.

In OR gate, the inputs A,B, C, etc., produce the output as A+B+C+ etc. The symbol and the truth table of two input OR gate are shown in the Figure 8.12.



A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

Figure 8.12 Symbol and Truth Table of OR Gate

A two input OR gate contains two input signals and only one output signal.

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